

FULL TECHNICAL DOCUMENTATION

Project: Asset Defender

1. Setup and Runtime

- Root scripts: npm run dev, npm run dev:client, npm run dev:server, npm run regression:profile-scores
- Server default port: 5000
- Database location: server/database/app.db

2. Directory Guide

client/: React UI and extension assets
server/: Express API and DB initialization
server/ml/: model training, prediction, and evaluation scripts
messagedatasets/ and profiledataset/: training data

3. Frontend Documentation

Core pages include Login, Signup, Dashboard, AnalyzePage, AnalysisResultPage, HistoryPage, and AdminDashboard. AnalyzePage supports scraping and manual content.

4. Backend Documentation

Express routes implement authentication, admin analytics, dashboard data, scraping trigger, and analysis persistence. JWT middleware protects user-scoped and admin-scoped resources.

5. ML Documentation

Training scripts: server/ml/train_model.py and server/ml/train_message_models.py. Evaluation script: server/ml/evaluation.py. Artifacts saved as .pkl files for inference.

6. Heuristic Engine Documentation

Profile heuristics inspect follower ratios, engagement, profile completeness, risky bio terms, suspicious username patterns, and content behavior. Message heuristics inspect urgency, manipulation phrases, scam markers, repetition, and burst patterns.

7. Data Model Documentation

users table stores credentials and account role. analysis tables store feature snapshots and risk components. final_predictions stores final risk and explanation payload.

8. Operational Documentation

Logs are available via server.log and server.err.log. Admin endpoints provide user list, aggregated stats, and analysis views for governance.

9. Extension Documentation

Manifest V3 scripts in client/public include popup, background, and Instagram analyzer logic for profile/message extraction and UI interaction.

10. Known Constraints

No full CI/CD pipeline in repository, limited automated test coverage, and legal policy documents for scraping should be formalized before production rollout.